Electric Vehicle Management System Using Artificial Intelligence and Machine Learning

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid adoption of electric vehicles (EVs) is transforming modern transportation by reducing carbon emissions and promoting sustainable mobility. However, many EV users face difficulties in interpreting system alerts, battery warnings, and technical fault notifications generated by vehicle monitoring systems. Existing EV dashboards typically display crude diagnostic codes or simple warnings without providing clear explanations or recommended actions. As a result, users often struggle to understand the severity of problems or how to respond appropriately, which can lead to unsafe driving decisions, rapid battery degradation, or costly maintenance.

A major research gap exists in current EV management systems: the lack of intelligent, user-centric diagnostic support that can translate complex technical fault information into understandable guidance. Most existing solutions focus mainly on monitoring battery metrics and charging statistics, but do not provide context-aware support to help users make informed decisions in unusual vehicle situations.

To bridge this gap, this research proposes an intelligent EV problem diagnosis system with a context-aware do-and-don't-do engine. The system will analyze EV sensor data, battery status information, and system alerts to identify anomalies and classify faults based on severity levels. Machine learning and deep learning models will be used to identify unusual patterns in vehicle behavior, and a rule-based guidance engine combined with natural language processing will generate clear, human-readable recommendations for users. These recommendations will include actionable advice on actions to take or avoid when a specific EV problem occurs.

The proposed system will use technologies such as Python-based machine learning frameworks, deep learning models for anomaly detection, and web-based dashboards for real-time warning visualization. The expected outcome is an intelligent decision-support mechanism that improves EV reliability, enhances user understanding of vehicle health conditions, and supports proactive maintenance practices.

By improving diagnostic transparency and providing actionable guidance, the system can increase user confidence in EV technology, reduce operational risks, and support the widespread adoption of sustainable transportation solutions.

Keywords: Electric Vehicles, Fault Diagnosis, Machine Learning, Context-Aware Guidance, Decision Support System.

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LIST OF ABBREVIATIONS

Abbreviation	Full form
AI	Artificial Intelligence
API	Application Programming Interface
BMS	Battery Management System
CNN	Convolutional Neural Network
DB	Database
DL	Deep Learning
EV	Electric Vehicle
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
KPI	Key Performance Indicator
LSTM	Long Short-Term Memory
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
OBD-II	On-Board Diagnostics II
REST	Representational State Transfer
RMSE	Root Mean Square Error
SoC	State of Charge
SoH	State of Health
UI	User Interface
UX	User Experience
V2G	Vehicle-to-Grid

INTRODUCTION

1.1 Background and Context

Electric vehicles (EVs) are rapidly becoming a key component of global sustainable transportation systems due to their ability to reduce greenhouse gas emissions and dependence on fossil fuels. With increasing environmental concerns and supportive government policies, the global EV market has experienced significant growth in recent years. According to international energy reports, EV sales exceeded 14 million units worldwide in 2023, accounting for a rapidly growing share of the global automobile market. This rapid adoption has created new challenges related to EV operation, maintenance, and user awareness of vehicle health conditions.

Modern EVs are equipped with multiple sensors and onboard diagnostic systems that continuously monitor battery health, charging conditions, motor performance, and other operational parameters. These systems generate various alerts and error codes when abnormal conditions are detected. However, most EV monitoring systems present these alerts as technical notifications or raw diagnostic codes that are difficult for average users to interpret. As a result, many EV owners struggle to understand the meaning of these warnings or determine what actions they should take when problems occur.

This issue affects several stakeholders, including EV owners, manufacturers, service providers, and charging infrastructure operators. EV users may ignore important alerts due to lack of understanding, which can lead to battery degradation, system failures, or unsafe driving conditions. Meanwhile, service providers may experience increased maintenance demand due to delayed fault reporting, and manufacturers may face reduced customer satisfaction due to unclear diagnostic information.

Existing EV monitoring platforms primarily focus on displaying basic information such as battery percentage, driving range, charging status, and error alerts. Although some systems provide diagnostic capabilities, they rarely offer clear explanations or actionable recommendations for users. Most platforms rely on simple rule-based alerts without intelligent interpretation of the context in which the problem occurs.

To address these limitations, this research proposes an **Intelligent EV Problem Diagnosis System with a Context-Aware Do's and Don'ts Engine**. The system will analyze EV operational data and sensor information to detect abnormal conditions and classify faults using machine learning techniques. In addition, a context-aware guidance engine will translate technical diagnostic results into simple, understandable recommendations for users. These recommendations will inform drivers about appropriate actions to take or avoid when specific EV issues occur.

The proposed research also contributes to the **United Nations Sustainable Development Goal (SDG) 7: Affordable and Clean Energy** and **SDG 11: Sustainable Cities and Communities** by supporting efficient EV usage and improving the reliability of sustainable transportation technologies.

From a research perspective, this study aligns with the **Technology Integration and Management (TIM) research cluster**, which focuses on integrating intelligent technologies, data analytics, and decision-support systems to improve modern technological infrastructures and user experiences.

1.2 Literature Review

Recent research has explored various approaches for improving electric vehicle diagnostics, battery management, and predictive maintenance systems. Many studies focus on battery health monitoring and degradation prediction using machine learning and data-driven techniques.

Battery degradation modeling has been widely studied in EV research. Several researchers have developed predictive models that estimate battery health based on charging cycles, temperature variations, and operational conditions. These models help identify long-term battery performance trends and support preventive maintenance strategies.

Fault diagnosis techniques have also been explored in EV systems using machine learning and data-driven analytics. Researchers have applied algorithms such as neural networks, support vector machines, and random forest classifiers to detect abnormal patterns in battery voltage, temperature, and current signals. These methods can identify early signs of system failures and improve vehicle reliability.

Another area of research focuses on intelligent transportation systems and smart EV monitoring platforms. Many modern EV management systems provide real-time monitoring of battery status, driving performance, and charging behavior. These platforms allow users to track vehicle conditions through mobile applications or onboard dashboards.

In addition to research studies, several commercial EV platforms offer diagnostic features and vehicle monitoring services. For example, EV manufacturers provide mobile applications that display battery levels, charging status, and driving statistics. Some advanced EV systems also include remote diagnostics and service notifications to inform users when maintenance may be required.

Despite these advancements, most existing solutions focus primarily on technical diagnostics and data monitoring rather than user-oriented guidance. Many systems still display complex diagnostic codes that require technical knowledge to interpret. Furthermore, current EV platforms rarely provide clear instructions or recommendations to help users respond appropriately to vehicle faults.

As a result, there remains a significant gap between technical EV diagnostics and practical user support. Bridging this gap requires intelligent systems capable of translating complex diagnostic information into meaningful and actionable guidance for everyday users.

1.3 Research Gap

Although significant research has been conducted in EV battery monitoring, fault detection, and predictive maintenance, most existing solutions focus primarily on technical analysis rather than user-centered decision support. Current EV monitoring systems typically present raw error codes, warning alerts, or basic system notifications without explaining the underlying causes of the problem or recommending appropriate user actions.

This lack of contextual interpretation creates difficulties for EV users, particularly those without technical knowledge of vehicle systems. When drivers encounter warning messages or diagnostic alerts, they may not understand whether the issue is critical or minor, what actions should be taken immediately, or what behaviors should be avoided to prevent further damage. Consequently, users may ignore important alerts, continue driving under unsafe conditions, or follow inappropriate charging practices that accelerate battery degradation.

Existing research in EV diagnostics also tends to focus on individual components such as battery health prediction, charging optimization, or fault detection models. However, limited research has explored integrated systems that combine machine learning–based anomaly detection with context-aware guidance for end users. Most current solutions lack mechanisms to convert technical diagnostic outputs into simple, human-readable instructions.

Therefore, there is a clear need for an intelligent EV diagnostic support system that not only detects faults but also provides context-aware recommendations in the form of actionable **Do's and Don'ts**. Such a system can bridge the gap between complex technical data and practical user understanding.

The novelty of this research lies in integrating machine learning–based anomaly detection, fault severity classification, and rule-based contextual guidance into a unified decision-support engine. By transforming technical EV diagnostic data into understandable instructions, the proposed system aims to improve user awareness, enhance vehicle safety, and promote proactive maintenance practices. This approach represents a significant step towards more intelligent and user-centered EV management systems.

OBJECTIVE

2.1 Main Objective

To develop and evaluate an **AI-based Intelligent Electric Vehicle Problem Diagnosis System with a Context-Aware Do's and Don'ts Engine** that can detect abnormal vehicle conditions from EV sensor data, classify faults based on severity, and provide clear, actionable guidance to EV users in order to support safe and informed decision-making.

2.2 Specific Objectives

1. **To collect and preprocess EV operational data** such as battery status, temperature readings, charging behavior, and system alerts in order to create a dataset suitable for anomaly detection and fault diagnosis.
2. **To design and implement a machine learning–based anomaly detection model** (e.g., LSTM or Autoencoder) capable of identifying abnormal patterns in EV sensor and operational data.
3. **To develop a fault classification model** using machine learning algorithms such as Random Forest or XGBoost to categorize detected EV issues according to severity levels (e.g., minor, moderate, critical).
4. **To design a context-aware Do's and Don'ts guidance engine** using rule-based reasoning and natural language generation to translate technical diagnostic outputs into simple and understandable recommendations for EV users.
5. **To develop a user interface/dashboard** that displays EV alerts, diagnostic explanations, and recommended actions in a clear and user-friendly format.
6. **To evaluate the effectiveness and accuracy of the proposed system** using performance metrics such as accuracy, precision, recall, and user usability testing to determine its ability to support EV users in responding to vehicle problems.

3. METHODOLOGY

Research Approach

This research adopts a **Design Science Research (DSR)** approach combined with experimental evaluation to develop and validate an intelligent EV diagnostic support system. Design Science is suitable for this study because the objective is to design and implement a technological artifact that solves a real-world problem related to EV fault interpretation and user decision support. The research will involve designing a prototype system, implementing machine learning models for fault detection and classification, and evaluating the system's performance using experimental testing.

The research process includes several stages: problem identification, system design, model development, implementation, and evaluation. First, EV operational data and diagnostic information will be analyzed to identify patterns related to vehicle faults. Then, machine learning and deep learning techniques will be used to build models capable of detecting anomalies and classifying EV issues. Finally, a context-aware guidance engine will be designed to convert technical diagnostic outputs into clear instructions for users.

Technical Methods and Algorithms

The proposed system will consist of several analytical modules that work together to detect, classify, and explain EV problems.

Anomaly Detection

Deep learning models such as **Long Short-Term Memory (LSTM) networks** or **Autoencoders** will be used to detect abnormal patterns in EV operational data. These models are suitable for time-series data such as battery voltage, temperature variations, and charging patterns because they can capture temporal dependencies and identify deviations from normal behavior.

Fault Classification

After anomalies are detected, machine learning algorithms such as **Random Forest** and **XGBoost** will be used to classify EV faults into severity levels (e.g., low, moderate, critical). These algorithms are chosen because they provide high classification accuracy and can handle complex feature relationships.

Context-Aware Guidance Engine

A **rule-based reasoning system** combined with **Natural Language Processing (NLP)** will be used to generate clear and understandable guidance for EV users. The system will map detected faults to recommended actions, producing instructions in the form of "Do's and Don'ts." This approach ensures that users receive actionable recommendations rather than raw technical error codes.

Tools and Platforms with Justification

Several software tools and platforms will be used to implement and evaluate the proposed system.

Python will be used as the primary programming language because it provides strong support for machine learning and data analysis.

Machine learning frameworks such as **TensorFlow**, **PyTorch**, and **Scikit-learn** will be used to implement anomaly detection and classification models due to their flexibility and extensive libraries for data science.

A **MongoDB database** will be used to store EV operational data, alerts, and diagnostic results because it supports scalable and flexible data storage for large datasets.

For system integration and backend services, **Node.js** will be used to develop APIs that connect the machine learning modules with the user interface.

A **React-based web dashboard** will be developed to display alerts, explanations, and recommended actions for EV users in a clear and interactive format.

These technologies are widely used in modern intelligent systems and allow efficient development, scalability, and real-time data processing.

Data Collection and Ethical Considerations

The proposed system will use EV operational data such as battery voltage levels, temperature readings, charging behavior, and diagnostic alerts. Data may be obtained through publicly available EV datasets, simulated datasets, or voluntarily shared data collected through a prototype mobile or web application.

To protect user privacy, all collected data will be **anonymized**, ensuring that no personally identifiable information is stored or processed. Participation in any data collection process will be voluntary and based on informed consent. Secure data storage and access control mechanisms will also be implemented to ensure confidentiality and prevent unauthorized access.

Ethical considerations will follow institutional research guidelines to ensure that the study maintains transparency, data protection, and responsible research practices.

Validation Strategy and Performance Metrics

The proposed system will be evaluated using both **technical performance metrics** and **usability assessment methods**.

For anomaly detection and fault classification models, the following evaluation metrics will be used:

- **Accuracy** to measure the overall correctness of predictions
- **Precision** to evaluate the proportion of correctly identified fault cases
- **Recall** to measure the system's ability to detect actual faults
- **F1-score** to provide a balanced evaluation of precision and recall

In addition to model performance evaluation, the usability of the context-aware guidance system will be assessed through **user testing and feedback surveys**. These tests will evaluate whether the generated Do's and Don'ts instructions are clear, understandable, and helpful for EV users when responding to vehicle alerts.

HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of Proposed Solution and Its Components

The proposed system is an **Intelligent Electric Vehicle Problem Diagnosis and Context-Aware Guidance System** designed to assist EV users in understanding vehicle alerts and responding appropriately to abnormal conditions. The architecture integrates data collection, machine learning analysis, and a decision support engine that generates clear instructions for users.

The system consists of several main components that work together to process EV operational data and provide meaningful guidance:

1. **Data Acquisition Module**
   This module collects EV operational data such as battery status, charging behavior, temperature readings, error codes, and other sensor information. Data can be obtained from onboard vehicle sensors, mobile applications, or simulated datasets.

2. **Data Processing and Storage Layer**
   The collected data is cleaned, transformed, and stored in a centralized database. This layer prepares the data for machine learning analysis by performing preprocessing tasks such as normalization, feature extraction, and filtering of invalid records.

3. **Anomaly Detection Module**
   This module uses deep learning models to analyze EV operational data and detect abnormal patterns. It identifies unusual behavior in battery performance, charging activity, or temperature fluctuations that may indicate potential system faults.

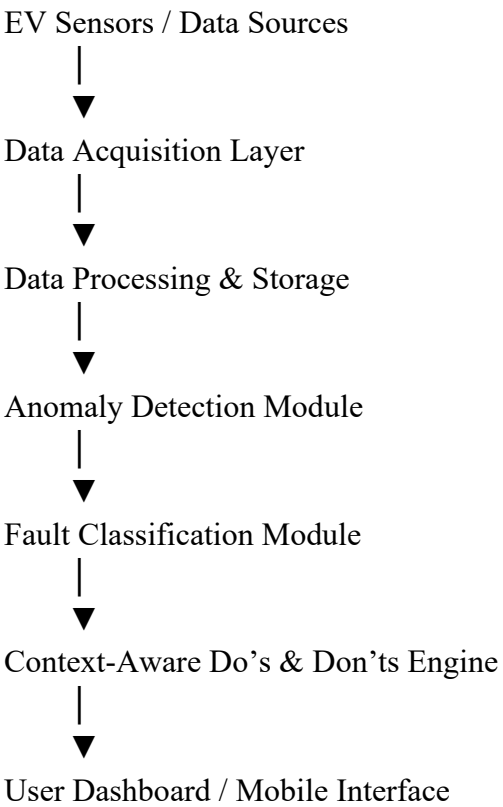
4. **Fault Classification Module**
   After detecting anomalies, machine learning algorithms classify the detected issues according to severity levels such as minor, moderate, or critical faults. This classification helps determine the urgency of the problem.

5. **Context-Aware Do's and Don'ts Engine**
   This is the **core component of the research**. It translates technical diagnostic results into understandable instructions for EV users. The engine uses rule-based reasoning and contextual analysis to generate actionable recommendations that guide users on what actions they should take or avoid.

6. **User Interface and Visualization Layer**
   The system presents diagnostic alerts, explanations, and recommended actions through a web or mobile dashboard. Visualizations help users easily understand the vehicle condition and respond accordingly.

System Diagram of Components

A simplified high-level architecture diagram can include the following layers:



This layered structure ensures a clear separation between data processing, intelligent analysis, and user interaction.

Individual Component and Its Parts

The **Intelligent EV Problem Diagnosis with Context-Aware Do's & Don'ts Engine** consists of several internal subcomponents:

1. **Alert Interpretation Module**
   Interprets incoming EV error codes and sensor alerts.
2. **Severity Assessment Module**
   Determines how critical the detected issue is based on machine learning classification results.
3. **Knowledge Base**
   Stores predefined rules, safety guidelines, and EV operational recommendations.
4. **Recommendation Generator**
   Produces contextual guidance in the form of Do's and Don'ts instructions.
5. **Explanation Generator**
   Converts technical information into human-readable explanations using simple language.

Component Interactions

Each module in the system interacts sequentially to process data and generate recommendations.

First, EV operational data is collected and stored in the data processing layer. The anomaly detection module analyzes this data to identify unusual patterns. If abnormal behavior is detected, the fault classification module determines the type and severity of the issue. The classification results are then passed to the context-aware guidance engine, which references the knowledge base to generate appropriate recommendations. Finally, these results are displayed to users through the dashboard interface.

This interaction allows the system to transform raw technical data into actionable insights for users.

Data Flow Explanation

The data flow within the system follows a structured pipeline:

1. EV sensor data and operational logs are collected.
2. Data is cleaned and stored in the central database.
3. Machine learning models analyze the data for anomalies.
4. Detected anomalies are classified into fault categories.
5. The context-aware engine generates recommendations.

6. The system displays alerts and guidance to the user interface.

This process enables real-time monitoring and intelligent decision support for EV users.

Scalability and Security Considerations

To ensure scalability, the system architecture supports modular integration and cloud-based deployment. The machine learning modules can be deployed as independent services, allowing the system to handle increasing volumes of EV data as the number of users grows. Cloud infrastructure and distributed databases can be used to support large-scale data processing and real-time analytics.

Security is also an important consideration in the system design. Sensitive data collected from EV users will be anonymized to protect user privacy. Secure communication protocols such as encrypted APIs will be used to transmit data between system components. Access control mechanisms will restrict unauthorized access to the database and analytical modules.

By incorporating scalable architecture and strong security practices, the system can support reliable and secure EV diagnostic services for a large number of users.

USER REQUIREMENTS

Stakeholder Identification

The proposed system involves multiple stakeholders who interact with or benefit from the EV diagnostic and guidance platform.

1. Electric Vehicle Owners / Drivers
EV drivers are the primary users of the system. They rely on the platform to understand vehicle alerts, detect potential problems, and receive clear instructions on what actions should be taken or avoided when an issue occurs.

2. EV Service and Maintenance Providers
Service centers and technicians can use diagnostic insights generated by the system to identify vehicle problems more efficiently and provide accurate maintenance services.

3. Electric Vehicle Manufacturers
Manufacturers can use aggregated diagnostic insights and failure patterns to understand common operational issues, improve vehicle design, and enhance after-sales service quality.

4. Charging Infrastructure Operators
Charging network operators may benefit from understanding charging-related faults and battery stress patterns, which can help improve charging station reliability and user support.

Functional Requirements

1. The system shall collect EV operational data such as battery status, temperature readings, charging events, and diagnostic alerts.
2. The system shall analyze collected EV data to detect abnormal behavior using anomaly detection techniques.
3. The system shall classify detected EV problems into severity categories such as minor, moderate, and critical faults.
4. The system shall interpret EV error codes and sensor alerts to determine the underlying issue.
5. The system shall generate context-aware recommendations in the form of **Do's and Don'ts** to guide users on appropriate actions.
6. The system shall provide clear explanations of detected problems in a human-readable format.
7. The system shall display diagnostic alerts, explanations, and recommended actions through a user-friendly dashboard or mobile interface.
8. The system shall allow users to view historical alerts and vehicle health information.

Non-Functional Requirements

1. Security
The system must ensure secure data storage and transmission. User data and vehicle information must be protected through encryption and access control mechanisms.

2. Performance
The system should process incoming EV data and generate diagnostic results within a reasonable time frame to support near real-time alerts.

3. Usability
The user interface should be simple, clear, and easy to understand so that non-technical EV users can easily interpret alerts and recommendations.

4. Scalability
The architecture should support the addition of new users and increased data volumes without significant degradation in performance.

5. Reliability
The system should maintain consistent operation and accurately detect EV faults with minimal errors.

6. Maintainability

The system should allow easy updates to machine learning models, rule-based knowledge bases, and software components as new EV technologies and diagnostic rules emerge

COMMERCIALIZATION PLAN

Target Market and Customer Persona

The proposed system targets the rapidly growing electric vehicle ecosystem, which includes individual EV owners, vehicle manufacturers, service providers, and charging infrastructure operators.

The **primary target market** is electric vehicle owners who require better understanding of vehicle alerts and guidance when technical issues occur. Many EV users lack technical knowledge about battery systems, charging conditions, and vehicle diagnostics, creating demand for intelligent tools that simplify vehicle monitoring and troubleshooting.

The **secondary target market** includes EV service centers and fleet operators who manage multiple electric vehicles and require efficient tools for identifying faults and improving maintenance efficiency.

A typical **customer persona** would be an EV owner who frequently uses a mobile application to monitor vehicle performance. This user values clear explanations of vehicle alerts and actionable recommendations that help prevent battery damage, reduce maintenance costs, and improve driving safety.

Value Proposition

The proposed system offers a unique value proposition by transforming complex EV diagnostic data into clear, actionable guidance. Instead of presenting technical error codes that are difficult to interpret, the system provides contextual explanations and practical recommendations in the form of Do’s and Don’ts.

Key benefits include improved vehicle reliability, reduced battery degradation, enhanced driver safety, and better understanding of EV operational conditions. The system helps users make informed decisions when vehicle issues arise, which can reduce unnecessary service visits and improve overall EV ownership experience.

Revenue Model

The system can adopt a **Software-as-a-Service (SaaS)** business model. Under this model, EV users or service providers subscribe to the platform to access advanced diagnostic features and intelligent recommendations.

Revenue may be generated through multiple channels, including:

- Monthly or annual subscription plans for EV users
- Licensing the technology to EV manufacturers or service providers
- Premium analytics services for fleet operators
- Integration services for EV mobile applications or vehicle dashboards

This diversified revenue approach allows the platform to scale as EV adoption increases.

Cost Estimation

The main costs associated with developing and deploying the system include software development, cloud infrastructure, and data processing resources.

Initial development costs may involve expenses related to machine learning model development, system integration, and interface design. Operational costs will include cloud storage, server hosting, database management, and continuous system maintenance.

Additional costs may arise from data acquisition, system testing, and periodic updates to maintain compatibility with evolving EV technologies.

Pricing Strategy

A **tiered pricing strategy** can be adopted to accommodate different types of users.

A basic plan may provide fundamental vehicle diagnostics and alert explanations at a low cost or as a free service. A premium plan can offer advanced features such as predictive fault detection, personalized recommendations, historical performance analysis, and detailed battery health reports.

Fleet operators and service providers may be offered enterprise-level packages with additional analytics and integration capabilities.

This flexible pricing model encourages wider adoption while generating sustainable revenue for system maintenance and expansion.

Competitive Advantage

The proposed system provides several competitive advantages compared to existing EV monitoring platforms.

First, it introduces a **context-aware guidance engine** that translates technical EV diagnostic data into clear instructions for users. Most current systems only display alerts or error codes without explaining what actions should be taken.

Second, the integration of machine learning–based anomaly detection with rule-based recommendations enables the system to provide proactive decision support rather than simple reactive monitoring.

Third, the platform focuses on **user-centered design**, ensuring that complex technical information is presented in a simple and understandable format. This improves user trust and engagement with EV technology.

Intellectual Property Considerations

The research may result in novel algorithms, system architectures, and decision-support mechanisms related to EV diagnostics and context-aware guidance. These innovations may be eligible for intellectual property protection.

Potential intellectual property considerations include software copyrights for the developed platform, proprietary machine learning models, and unique rule-based knowledge frameworks used for generating contextual recommendations.

If the system demonstrates strong commercial potential, patent protection or licensing agreements may be considered to protect the underlying technology and support future commercialization opportunities.

BUDGET AND JUSTIFICATION

The proposed research requires several resources including hardware, software tools, cloud infrastructure, and human effort. The estimated budget focuses on supporting system development, machine learning model implementation, and deployment of the EV diagnostic platform.

Hardware Costs

Table 1- Hardware Costs

Item	Estimated Cost (LKR)	Description
Laptop / Workstation	250,000	A high-performance computer with sufficient processing power, memory, and storage is required for machine learning model development, data processing, and system implementation.
External Storage Device	20,000	Used for backup storage of datasets, model checkpoints, and experimental results.
Testing Devices (Mobile / IoT Simulation)	40,000	Used for testing the user interface and simulating EV data inputs.

Total Estimated Hardware Cost: 310,000 LKR

Justification:
Machine learning models and data processing tasks require sufficient computing resources. A high-performance workstation enables efficient training of models, running simulations, and developing the prototype system.

Software Licenses

Table 2 - Software Licenses

Item	Estimated Cost (LKR)	Description
Development Tools (IDE, code management)	10,000	Tools used for coding, debugging, and version control management.
Data Analysis / Visualization Tools	10,000	Used for analyzing EV datasets and visualizing results.

Total Estimated Software Cost: 20,000 LKR

Justification:

Although many machine learning frameworks are open-source, certain development tools or plugins may require licenses to improve productivity and support efficient software development.

Cloud Services

Table 3 - Cloud Services

Item	Estimated Cost (LKR)	Description
Cloud Computing Services	30,000	Used for hosting the application, running machine learning models, and processing data.
Cloud Database Storage	15,000	Used to store EV sensor data, alerts, and processed results.
API Hosting and Backend Services	15,000	Supports system integration and real-time data communication between modules.

Total Estimated Cloud Cost: 60,000 LKR

Justification:

Cloud infrastructure is necessary to support scalable data storage, remote system access, and deployment of machine learning services. It enables the prototype system to function as an online platform.

Human Effort Cost

Table 4 - Human Effort Cost

Item	Estimated Cost (LKR)	Description
Research and Development Effort	100,000	Time spent on system design, machine learning model development, experimentation, and documentation.
System Development and Testing	60,000	Effort involved in implementing the application, integrating modules, and conducting testing and validation.

Total Estimated Human Effort Cost: 160,000 LKR

Justification:
Developing an intelligent EV diagnostic system requires significant research effort, algorithm development, data analysis, and system testing. These activities involve time investment and technical expertise.

Total Estimated Budget

Table 5 - Total Estimated Budget

Category	Estimated Cost (LKR)
Hardware	310,000
Software	20,000
Cloud Services	60,000
Human Effort	160,000

Total Estimated Project Cost: 550,000 LKR

WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the overall research project into smaller tasks assigned to each team member. This ensures efficient task management, proper timeline alignment, and balanced workload distribution across the research group. The project consists of four main components handled by different members of the research group.

Task Breakdown per Member

Table 6 - Task Breakdown per Member

Member	Component	Key Tasks
IT22157928	Post-Sale EV Performance Intelligence & Failure Prediction	Collect EV usage data, analyze performance trends, develop analytics models, identify failure patterns, build seller dashboard
IT22247490	Adaptive Battery Health Prediction	Monitor charging behavior, analyze driving patterns, build battery health prediction model, generate battery degradation forecasts
IT22134080	Intelligent EV Problem Diagnosis with Context-Aware Do's & Don'ts Engine	Develop anomaly detection model, classify EV faults, design rule-based guidance system, generate user recommendations
IT22071170	Predictive Charging & Breakdown Assistance	Predict charging station availability, develop charging recommendation system, design breakdown route assistance, implement emergency guidance

Timeline Alignment

The research tasks are distributed across the project timeline to ensure smooth progress and proper coordination between components.

Table 7 - Timeline Alignment

Phase	Activities
Phase 1	Literature review, problem analysis, requirement gathering
Phase 2	Dataset collection and preprocessing
Phase 3	Machine learning model development
Phase 4	System architecture design and module implementation
Phase 5	Integration of system components
Phase 6	Testing, validation, and performance evaluation
Phase 7	Documentation and final report preparation

Each component progresses through these phases simultaneously while maintaining integration points with the overall system.

Balanced Workload Distribution

The workload is distributed among team members based on the complexity and scope of each component.

- Each member is responsible for designing and implementing one core system module.
- Machine learning model development tasks are evenly distributed across components.
- System integration tasks are shared among all members to ensure compatibility between modules.
- Documentation, testing, and evaluation activities are collaboratively performed to maintain consistency across the project.

This structured distribution ensures that no single team member carries a disproportionate workload and that all components contribute equally to the overall Intelligent EV Management System.

GANTT CHART

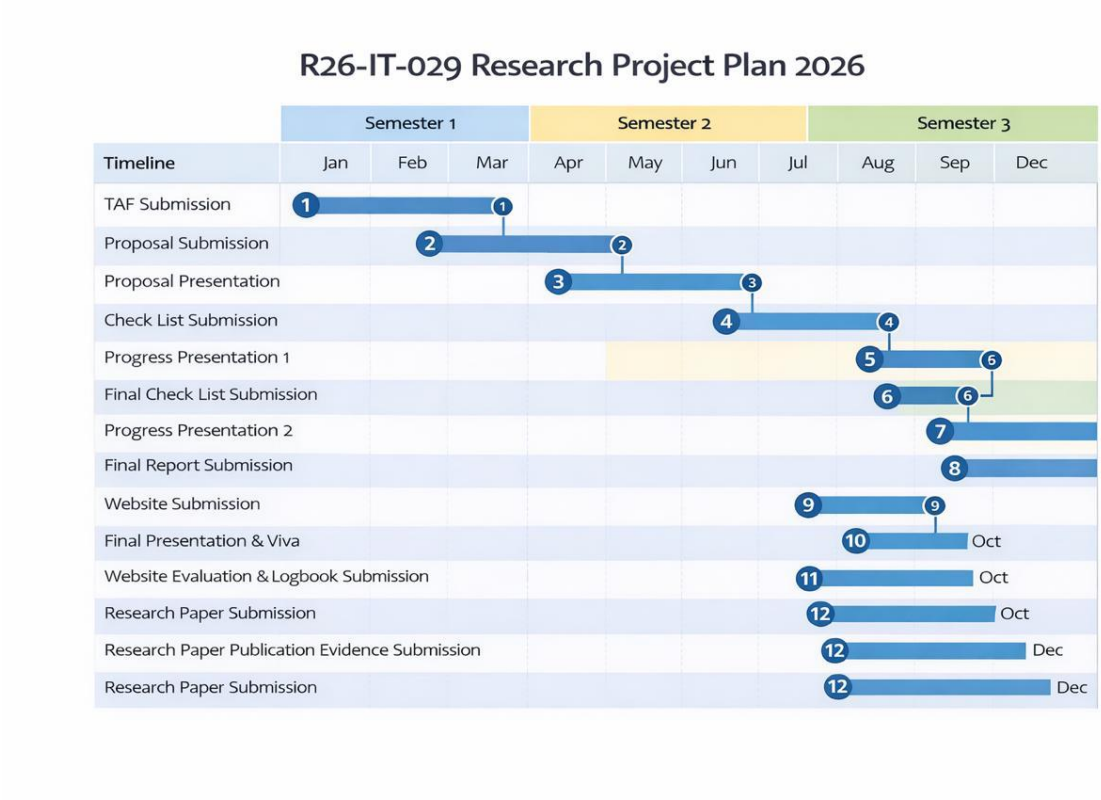


Figure 1 GANTT CHART

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APPENDICES

Appendix A – Survey Instruments

A survey will be conducted among electric vehicle users to understand common EV issues and user responses to vehicle alerts.

Sample Survey Questions

1. How long have you been using an electric vehicle?
2. Have you ever experienced unexpected EV system warnings?
3. How easy is it for you to understand EV error messages?
4. Have you experienced battery overheating or abnormal charging behavior?
5. Would you prefer a system that explains EV errors in simple instructions?
6. How often do you check EV battery health information?
7. What kind of guidance would you expect during EV failures?

The collected responses will help improve the design of the context-aware diagnostic system.

Appendix B – Detailed System Diagram

The proposed system consists of the following main components:

1. **EV Data Collection Layer**
    - Collects sensor data, battery status, and error logs.

2. **Fault Detection Module**
    - Uses machine learning models such as LSTM or Autoencoder to detect anomalies.

3. **Fault Classification Module**
    - Uses Random Forest or XGBoost to determine severity levels.

4. **Context-Aware Recommendation Engine**
    - Generates Do's and Don'ts instructions based on detected faults.

5. **User Interface Layer**
    - Displays alerts, guidance messages, and system recommendations.

Appendix C – Preliminary UI Mockups

The proposed user interface includes the following screens:

Dashboard Screen

- Battery health status
- Active alerts
- System recommendations

Fault Alert Screen

- Clear explanation of EV issue
- Do's and Don'ts instructions
- Emergency actions

History Screen

- Previous diagnostic alerts
- Battery performance trends
- Vehicle health reports



Figure 2- Preliminary UI Mockups

Appendix D – Risk Analysis

Table 8 - Risk Analysis

Risk	Description	Mitigation
Data Availability	Lack of real EV datasets	Use simulated datasets and public datasets
Model Accuracy	ML models may produce incorrect predictions	Perform extensive testing and validation
User Adoption	Users may not trust automated advice	Provide clear explanations and reliable recommendations
System Integration	Difficulty integrating different modules	Use modular architecture and APIs

Appendix E – Ethical Considerations

The research involves collecting EV usage data from participants. Ethical considerations include:

- Obtaining **informed consent** from participants.
- Ensuring **data anonymization** to protect user identity.
- Avoiding collection of sensitive personal data.
- Secure storage and handling of collected datasets.

Ethical approval will be obtained from the institutional review committee if required.